

Stationary Time Series Models

Chapter 3

Nicolas Cachanosky

Department of Economics and Finance

Outline

- The AR family
- The MA family
- The Wold decomposition
- ARMA models and identification
- Diagnostics
- ARMAX
- Forecasting

The AR family

Section 1

The AR family

- Why **autoregressive** models?
- The distributive-lag (DL) model
 - $y_t = \alpha + \mathbf{w}_0 \mathbf{X}_t + \underbrace{\mathbf{w}_1 \mathbf{X}_{t-1}}_{y_{t-1}} + \underbrace{\mathbf{w}_2 \mathbf{X}_{t-2}}_{y_{t-2}} + \dots$
 - The DL model has too many parameters
 - Use lags of y_t , which already includes contemporaneous and past values of $\mathbf{X}$
 - A model such as $y_t = \phi y_{t-1}$ is a parsimonious substitute for the DL model
 - The lagged dependent variable (y_t) acts as a sufficient statistic for all the past history of $\mathbf{X}$.

The AR family

- The **autoregressive** AR(1) process
 - $y_t = \phi y_{t-1} + c + \varepsilon_t; \quad \varepsilon_t \sim WN(0,1)$
 - Take y_t and subtract its own memory; what is left is a constant + a random shock
 - $\underbrace{y_t - \phi y_{t-1}}_{\text{explained}} = \underbrace{c + \varepsilon_t}_{\text{stochastic}}$
 - Remember the stability condition:
 - $|\phi| < 1 \rightarrow$ stable
 - $|\phi| = 1 \rightarrow$ unit-root process
 - $|\phi| > 1 \rightarrow$ unstable (explosive behavior)

The AR family

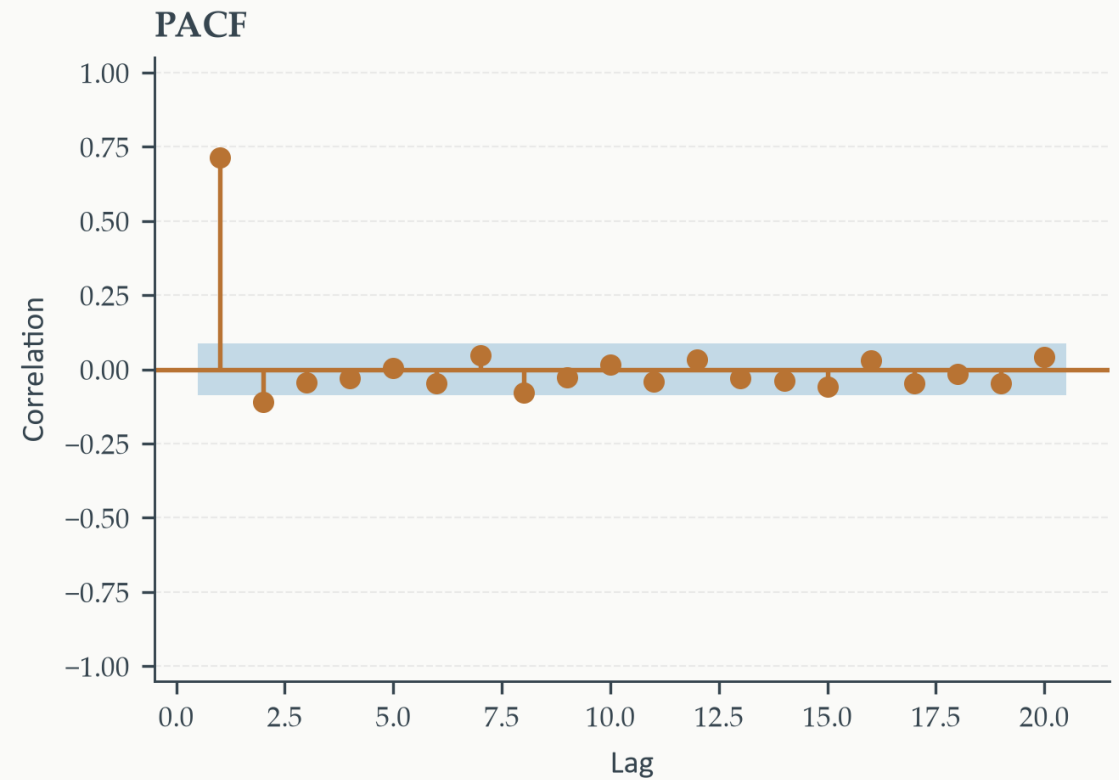
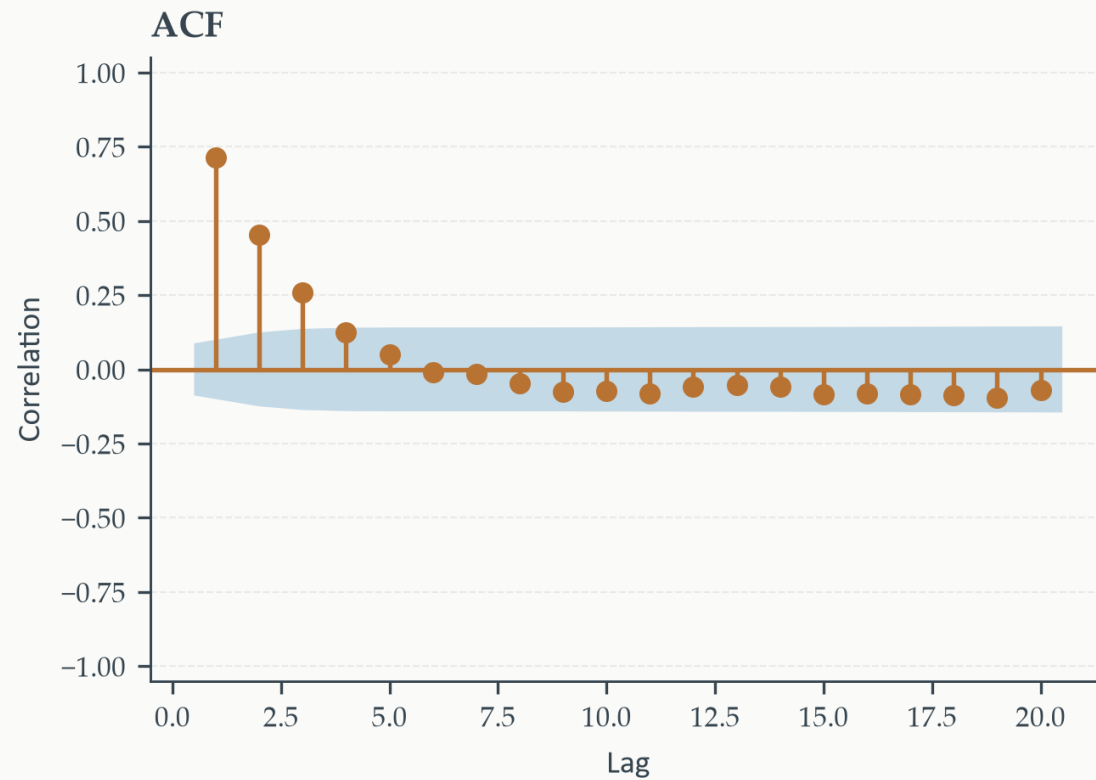
- From AR(1) to AR(p)
 - AR(1): $y_t = \phi_1 y_{t-1} + c + \varepsilon_t$
 - AR(2): $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + c + \varepsilon_t$
 - AR(3): $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + \phi_3 y_{t-3} + c + \varepsilon_t$
 - AR(p): $y_t = \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + c + \varepsilon_t$
 - AR(p): $y_t = \sum_{i=1}^p \phi_i y_{t-i} + c + \varepsilon_t$
 - AR(p): $y_t - \sum_{i=1}^p \phi_i y_{t-i} = c + \varepsilon_t$
 - AR(p): $\phi_p(L)y_t = c + \varepsilon_t$

The AR family

- AR(p) stability
 - An AR(p) process has p characteristic roots
 - Stability conditions
 - Stable: The modulus of **all** $|\lambda_1|, \dots, |\lambda_p| < 1$
 - Unit root: At least modulus of one $|\lambda_i| = 1$, all others must be $|\lambda_{j \neq i}| < 1$
 - Unstable: At least one modulus $|\lambda_i| > 1$

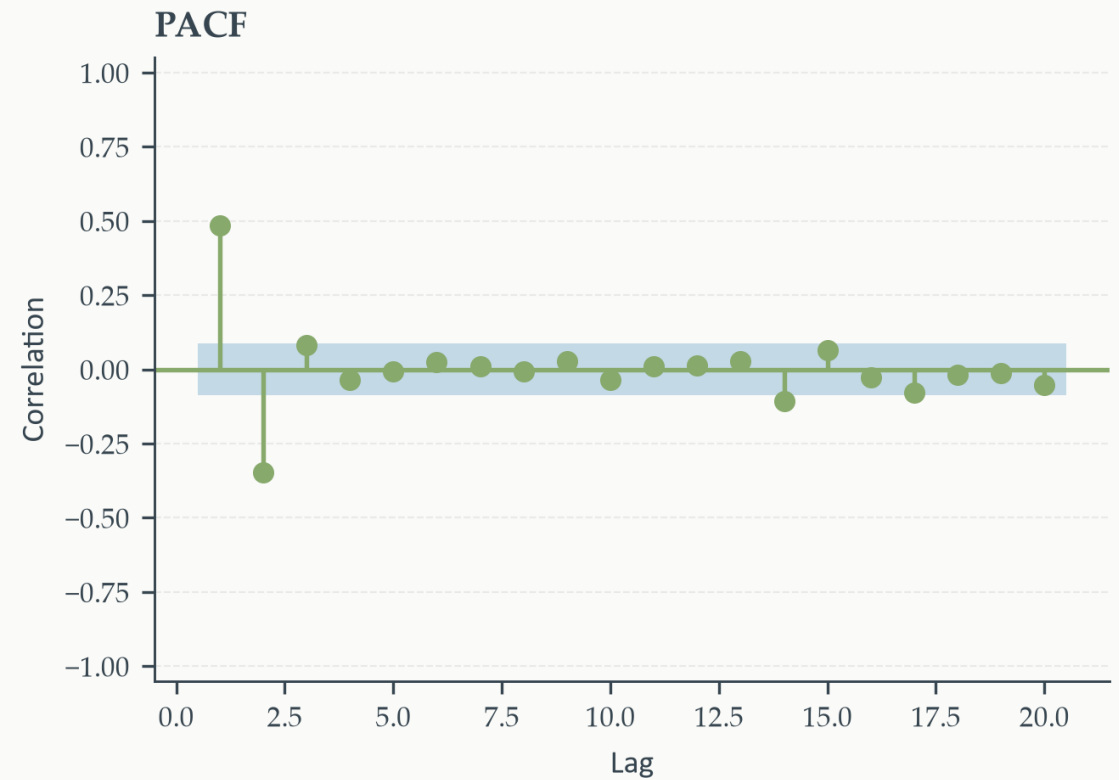
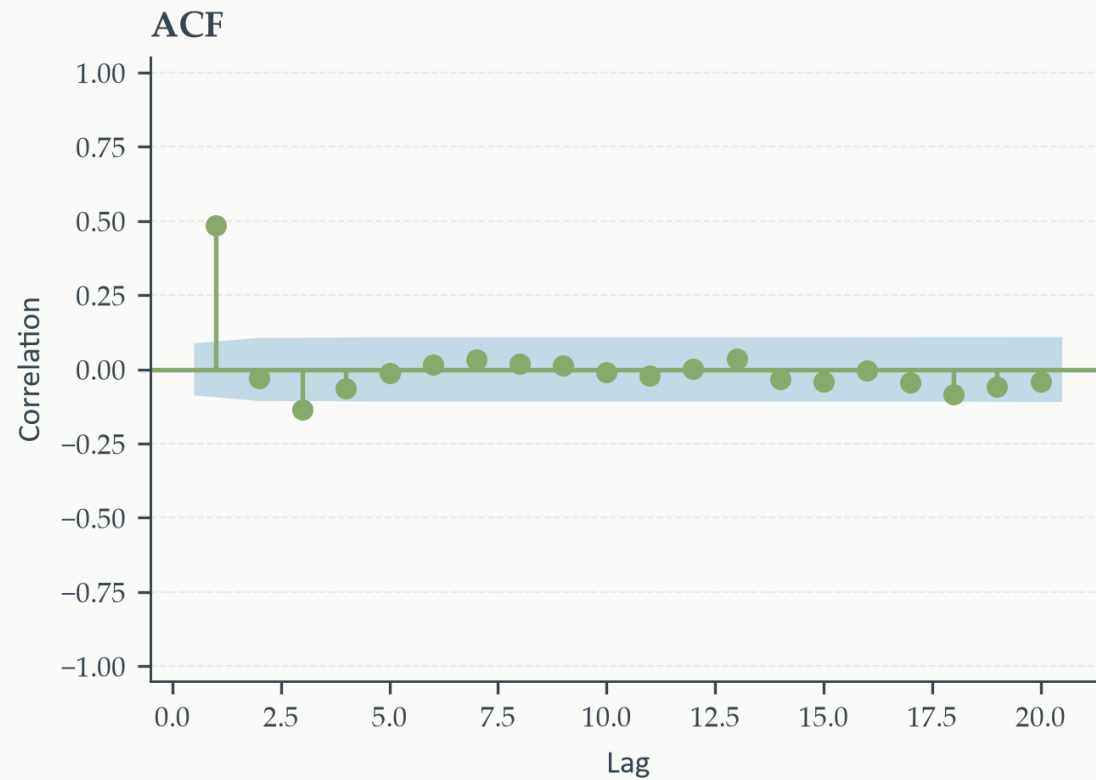
The AR family

AR(1): $\phi_1 = 0.7$



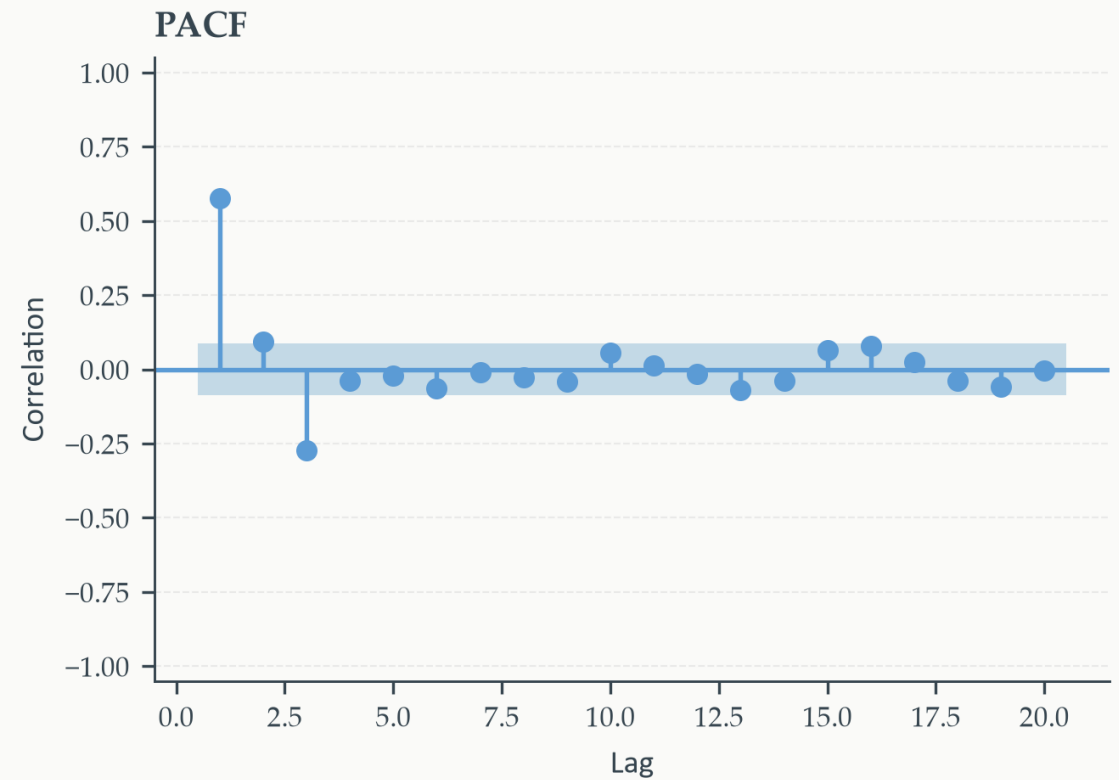
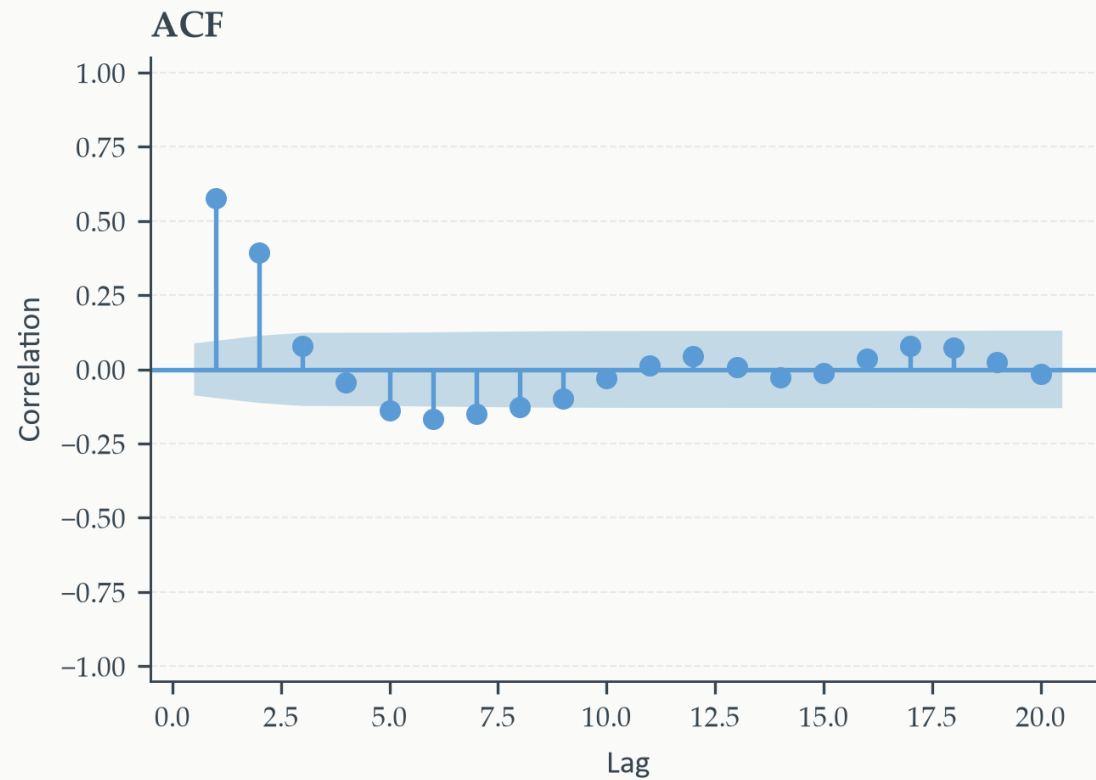
The AR family

AR(2): $\phi_1 = 0.6$, $\phi_2 = -0.3$



The AR family

AR(3): $\phi_1 = 0.5$, $\phi_2 = 0.2$, $\phi_3 = -0.25$



The AR family

- The Yule-Walker Equations: The AR(1) case
 - Step 1: Demean the series ($\tilde{y}_t = y_t - \bar{y}$):
 - $\tilde{y}_t = \phi_1 \tilde{y}_{t-1} + c + \varepsilon_t$
 - Step 2: Multiply by $\tilde{y}_{t-h}$ and take expectations
 - $\mathbb{E}[\tilde{y}_t \tilde{y}_{t-h}] = \phi_1 \mathbb{E}[\tilde{y}_{t-1} \tilde{y}_{t-h}] + \underbrace{\mathbb{E}[c \tilde{y}_{t-h}]}_0 + \underbrace{\mathbb{E}[\varepsilon_t \tilde{y}_{t-h}]}_0$
 - Step 3: You are left with covariances
 - $\gamma(h) = \phi_1 \gamma(h-1)$
 - Step 4: Normalize by $\gamma(0)$
 - $\frac{\gamma(h)}{\gamma(0)} = \rho(h) = \phi_1 \frac{\gamma(h-1)}{\gamma(0)} \rightarrow \text{if } h = 1 \rightarrow \rho(1) = \phi_1$

The AR family

- The Yule-Walker Equations: The AR(1) case (cont...)
 - Unroll h starting from zero, then:
 - $\gamma(h) = \phi_1 \gamma(h-1) = \phi_1^2 \gamma(h-2) = \dots = \phi_1^h \gamma(0)$
 - Normalize by $\gamma(0)$
 - $\rho(h) = \phi_1^h$

The AR family

- The Yule-Walker Equations: The AR(p) general case
 - $\mathbf{R}_{p \times p} \boldsymbol{\phi}_{p \times 1} = \boldsymbol{\rho}_{p \times 1}$, $R_{ij} = \rho(|i - j|) \rightarrow$ symmetric autocorrelation matrix
 - $\boldsymbol{\phi} = \mathbf{R}^{-1} \boldsymbol{\rho}$
 - Unique solution? Yes, when $\{y_t\}$ is stationary

$$\begin{pmatrix} \phi_1 \\ \phi_2 \\ \vdots \\ \phi_p \end{pmatrix} = \begin{pmatrix} \rho(0) & \rho(1) & \dots & \rho(p-1) \\ \rho(1) & \rho(0) & \dots & \rho(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ \rho(p-1) & \rho(p-2) & \dots & \rho(0) \end{pmatrix}^{-1} \begin{pmatrix} \rho(1) \\ \rho(2) \\ \vdots \\ \rho(p) \end{pmatrix}$$

The MA family

Section 2

The MA family

- AR captures one kind of memory, y on itself
- MA captures a different kind of memory, those related to past shocks not originated by $\{y_t\}$
 - A random shock can move y_{t-1} , and this will have an effect on y_t
 - But the movement of y_{t-1} is not “internal” to y , it is exogenous
 - The MA process captures the propagation of **random** shocks

The MA family

- The moving average MA(1) process
 - $y_t = \bar{y} + \varepsilon_t + \theta_1 \varepsilon_{t-1}; \quad \varepsilon_t \sim WN(0,1)$
 - No feedback from past levels of y , only from past shocks
 - This affects the variance of y
 - $\sigma_y^2 = Var(\bar{y} + \varepsilon_t + \theta_1 \varepsilon_{t-1}) = Var(\varepsilon) - \underbrace{2Cov(\varepsilon_t, \theta_1 \varepsilon_{t-1})}_0 + Var(\theta_1 \varepsilon_{t-1})$
 - $\sigma_y^2 = \sigma^2 + \theta_1^2 \sigma^2 = \sigma^2(1 + \theta_1^2) > \sigma^2$

The MA family

- The MA(q) process
 - MA(1): $\bar{y} + \varepsilon_t + \theta_1 \varepsilon_{t-1}$
 - MA(2): $\bar{y} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$
 - MA(3): $\bar{y} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2} + \theta_3 \varepsilon_{t-3}$
 - MA(q): $y_t = \bar{y} + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_q$
 - MA(q): $y_t = \bar{y} + \varepsilon_t + \sum_{i=1}^q \theta_i \varepsilon_{t-i}$
 - MA(q): $y_t - \bar{y} = \varepsilon_t + \sum_{i=1}^q \theta_i \varepsilon_{t-i}$
 - MA(q): $y_t - \bar{y} = \theta_q(L) \varepsilon_t$

The MA family

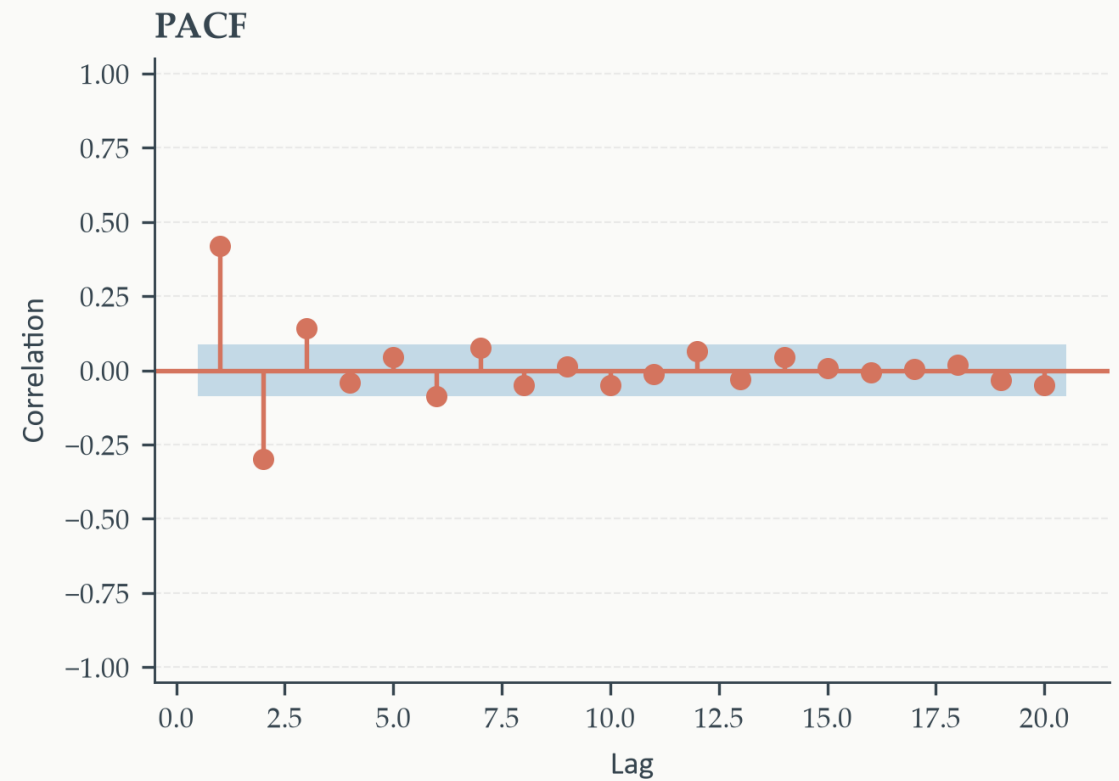
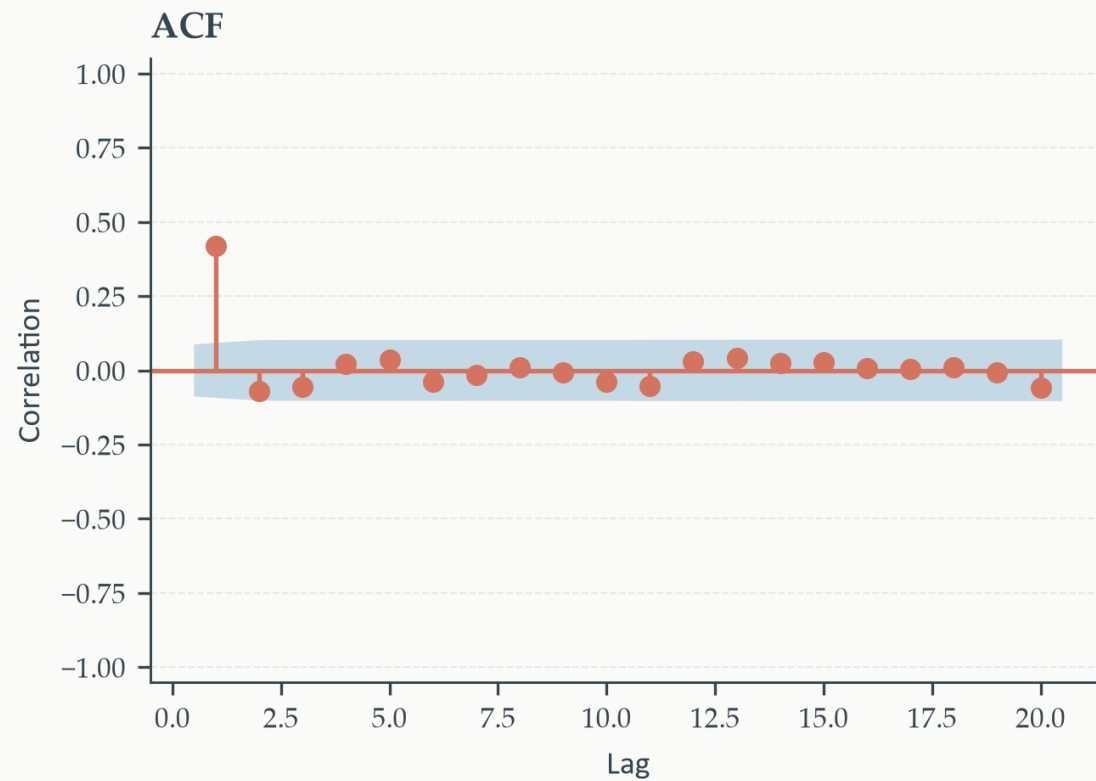
- Invertibility and Stationarity
 - Stationarity – condition of $\{y_t\}$
 - Does y_t have finite variance?
 - This is automatic in the MA(q) family because ε has finite variance
 - Invertibility — condition of $\{\varepsilon_t\}$
 - Can we recover ε from past observations of y_t ?
 - $y_t - \bar{y} = (1 + \theta_1 L)\varepsilon_t$
 - $\varepsilon_t = (1 + \theta_1 L)^{-1}(y_t - \bar{y})$; $(1 + \theta_1 L)^{-1} \rightarrow$ infinite series
 - $\varepsilon_t = (y_t - \bar{y}) - \theta_1(y_{t-1} - \bar{y}) + \theta_1^2(y_{t-2} - \bar{y}) - \dots$
 - $\varepsilon_t \xrightarrow{\text{converges to}} < \infty \text{ iff } |\theta_1| < 1$

The MA family

- The mirror image of the AR family
 - AR(p)
 - ACF gradually tails off
 - ACF cuts off at lag p
 - MA(q)
 - ACF cuts off at lag q
 - PACF gradually tails off

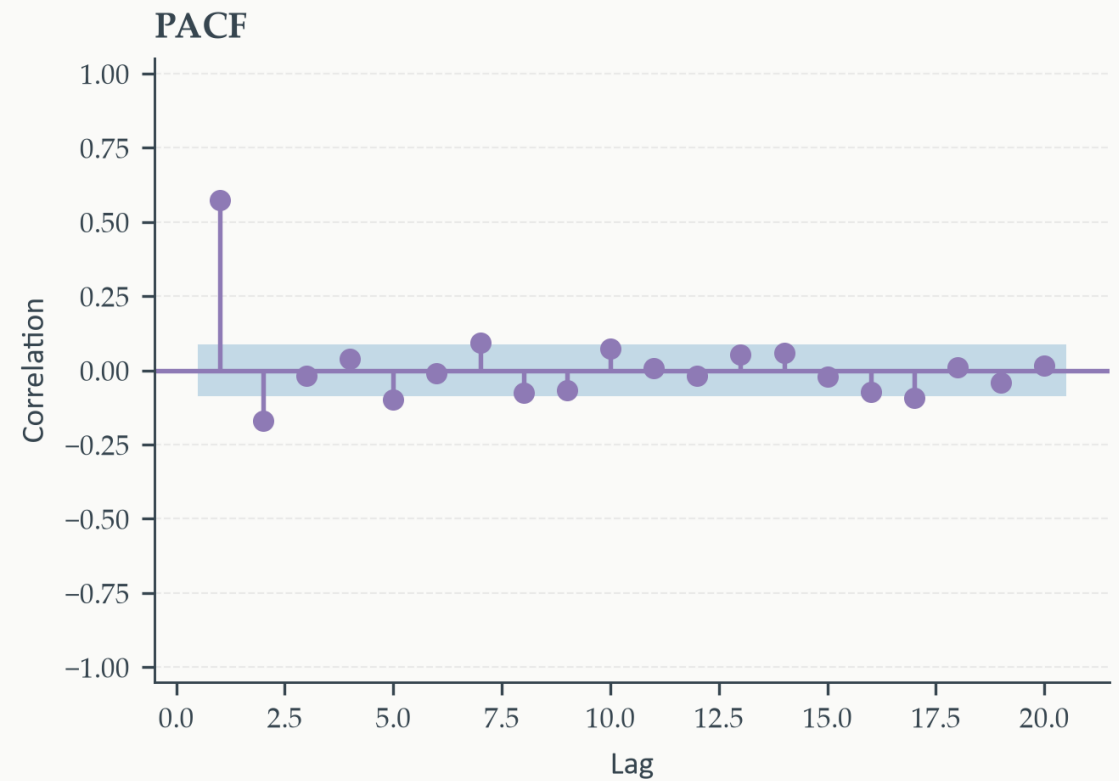
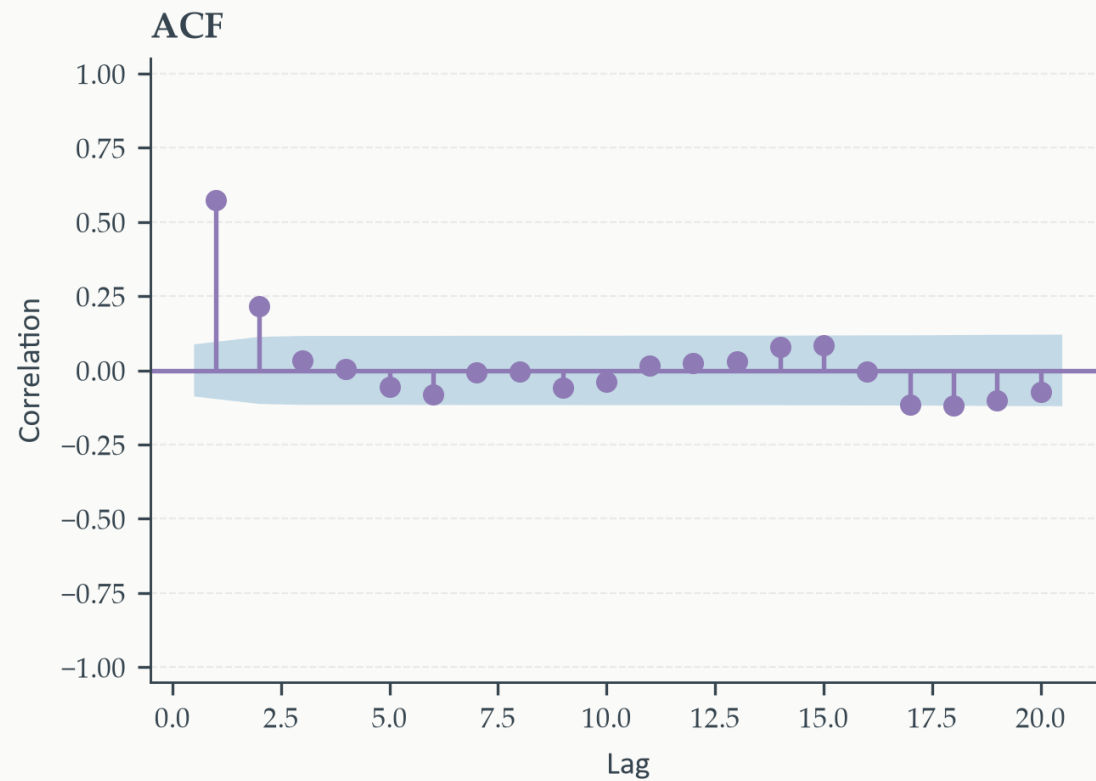
The MA family

MA(1): $\theta_1 = 0.7$



The MA family

MA(2): $\theta_1 = 0.6$, $\theta_2 = 0.3$



The MA family

- We already know why we care about stationarity
- Why should we care about invertibility?
 - Random shocks are **unobservable**; invertibility allows us to recover them from data
 - An invertible (non-invertible) MA process can (cannot) be approximated by a high-order AR process

The MA family

- How do we get $\hat{\varepsilon}_t$?
 - Iterative substitution
 - **Step 1:** Start with a large AR(p) process as a first guess; extract the residuals ($e_t \approx \varepsilon_t$)
 - **Step 2:** Use e_t as proxy of ε_t in the MA equation; re-estimate the MA parameters $\hat{\theta}$
 - **Step 3:** With updated $\hat{\theta}$, recompute ε_t
 - **Step 4:** Repeat Steps 2-3 until convergence (parameters stop changing)

The Wold decomposition

Section 3

The Wold decomposition

- Invertibility
 - A MA(q) process can be proxied by an AR(∞) process
- Wold decomposition
 - An AR(p) process can be represented by a MA(∞) process
 - $\phi_p(L)y_t = c + \varepsilon_t$
 - $y_t = \phi_p(L)^{-1}(c + \varepsilon_t) = \bar{y} + \varepsilon_t + (\theta_1\varepsilon_{t-1} + \theta_2\varepsilon_{t-2} + \cdots) = \bar{y} + \theta(L)\varepsilon_t$
 - θ_j : impulse response at horizon j — the effect on y_t of a shock j periods ago
 - If AR(p) is stationary, then $\theta_j \rightarrow 0$ with growing values of j

The Wold decomposition

- World decomposition theorem
 - Every covariance-stationary process $\{y_t\}$ with mean zero can be written as

$$y_t = \sum_{j=0}^{\infty} \theta_j \varepsilon_{t-j} + \eta_t$$

$y_t =$ linearly indeterministic + linearly deterministic

- Where:
 - $\theta_0 = 1, \sum \theta_j^2 < \infty, \{\varepsilon_t\} \sim WN(0, \sigma^2)$ are the **Wold innovations**
 - η_t is perfectly predictable from its own past and uncorrelated with every ε_t (typically removed when demeaning)

The Wold decomposition

- Why do we care about Wold decomposition?
 - Justifies MA models. A MA(q) model is a truncated Wold decomposition
 - Justifies AR approximation. An AR(p) model can be approximated by an MA($p < q$) model
 - Defines innovation. ε_t is the residual from the **best linear predictor** of y_t
 - Connects to IRFs. Every stationary $\{y_t\}$ has impulse responses, and these are captured by θ_j

The AR and MA families are not one subordinated to the other — they are **dual representations** of the same data. This duality becomes an important tool once we reach the ARMA model specification

ARMA models and identification

Section 4

ARMA models and identification

- An ARMA(p,q) model combines the AR(p) and MA(q) families into one model

$$y_t = \underbrace{(\phi_1 y_{t-1} + \cdots + \phi_p y_{t-p})}_{AR(p)} + c + \underbrace{(\varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q})}_{MA(q)}$$

$$y_t = \left(\sum_{i=1}^p \phi_i y_{t-i} \right) + c + \left(\varepsilon_t + \sum_{j=1}^q \theta_j \varepsilon_{t-j} \right)$$

$$\phi_p(L)y_t = \bar{y}^* + \phi_q(L)\varepsilon_t$$

$$\bar{y}^* \neq \bar{y} \rightarrow \bar{y}^* = (1 - \phi_1 - \cdots - \phi_p)\bar{y}$$

ARMA models and identification

- The parsimony trade-off
 - An ARMA(p, q) process requires less coefficients than an AR(p) or MA(q) model
 - ARMA(p, q) implies $(p + q) < p$ only and $(p + q) < q$ only
- The AR component absorbs the slowly decaying part of the autocorrelation, and the MA component shapes the initial decay

ARMA models and identification

- The cancellation problem
 - What happens if the $AR(p)$ and $MA(q)$ processes share a characteristic root?
 - They “cancel out”, producing a lower-order ARMA model
 - The underlying issue: Redundant parameters
 - Reduce the model order until there are no common roots left

ARMA models and identification

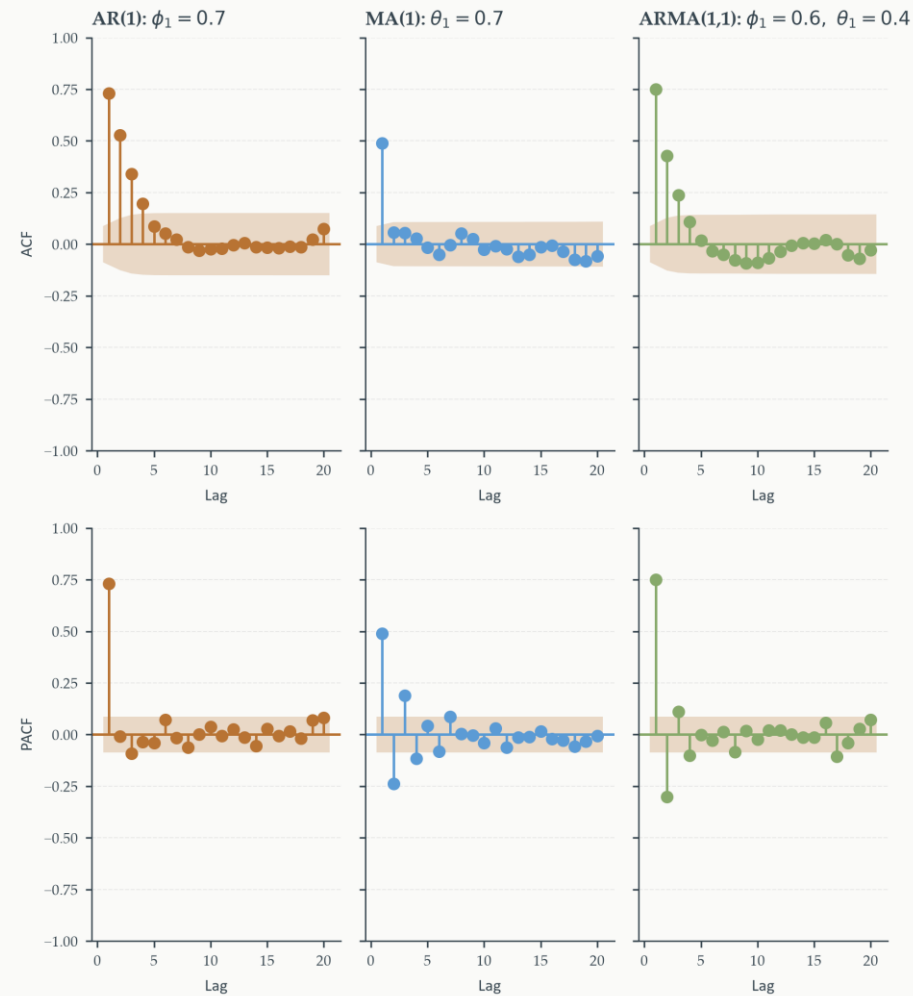
Model	ACF	PACF
$AR(p)$	Tails off	Cuts off after lag p
$MA(q)$	Cuts off after lag q	Tails off
$ARMA(p, q)$	Tails off	Tails off
White noise	Zero at all lags	Zero at all lags

ARMA models and identification

- The Box-Jenkins Workflow
 - **Step 0:** Check for stationarity (next chapter)
 - **Step 1:** Initial identification. Inspect ACF/PACF and select the best visual candidate for p and q
 - **Step 2:** Fit the model (estimate ϕ and θ) by maximum likelihood
 - **Step 3:** Model diagnostics. Check if residuals behave as WN. If they don't, return to Step 1
 - Collect candidate models by iterating Steps 1 through 3
 - **Step 4:** Model selection. Choose your best model, typically by information criteria and parsimony
 - **Step 5:** Forecasting! 😊

ARMA models and identification

ACF and PACF: AR(1), MA(1), ARMA(1,1) — $T = 500$



ARMA models and identification

- Maximum likelihood: The idea
 - Assume a probability distribution
 - What are the parameters (ϕ, θ) more likely to come from the assumed distribution **given** the observed data
- Take the log of a normal distribution, and minimize the likelihood ℓ

$$\min_{\{\phi, \theta\}} \ell(\phi, \theta, \bar{y}, \sigma^2) = -\frac{T}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{t=1}^T \varepsilon_t^2(\phi, \theta)$$

ARMA models and identification

- Information criteria
 - Penalize log-likelihood by the number of estimated parameters
 - $IC = -2\hat{\ell} + \text{penalty}(k, T)$
 - k : estimated parameters
 - First term: Rewards fit
 - Second term: Penalizes complexity — large k
 - The lower number the better!
 - $IC = 1$ is better than $IC = 5$
 - $IC = -2$ is better than $IC = 1$
 - $IC = -5$ is better than $IC = -2$

ARMA models and identification

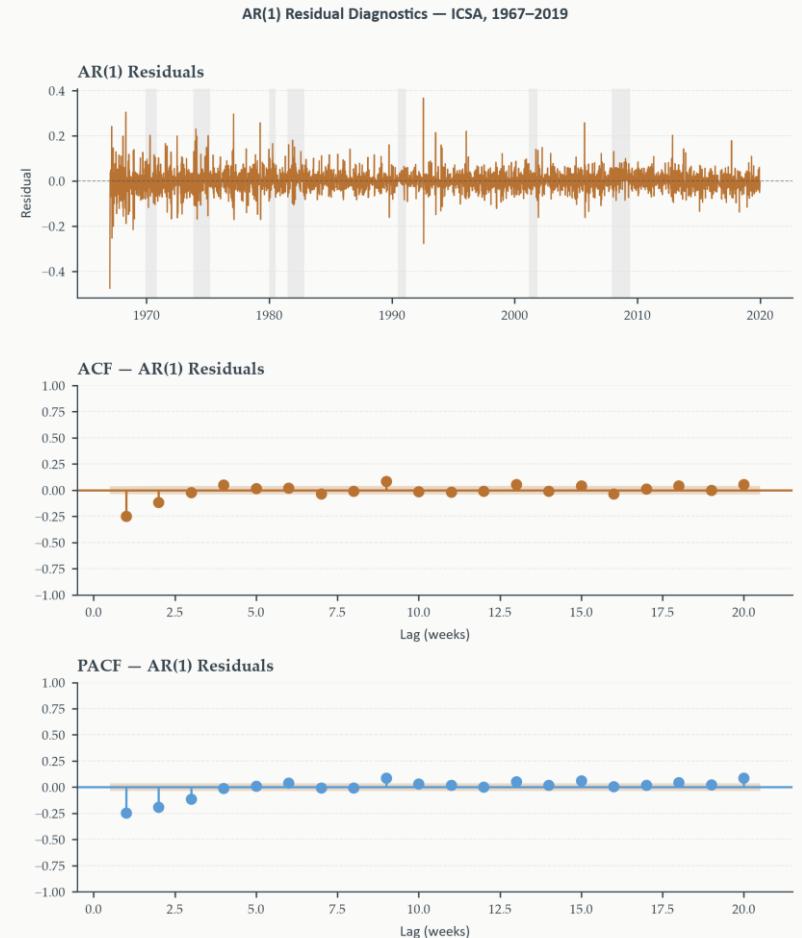
- Information criteria (cont...)

Criterion	Formula	Penalty per parameter	Best for ...
AIC	$-2\hat{\ell} + 2k$	2	Forecasting — tolerates large models
BIC	$-2\hat{\ell} + k \ln(T)$	$\ln(T)$	Model-order inference
HQ	$-2\hat{\ell}k \ln(\ln(T))$	$2 \ln(\ln(T))$	Compromise

Why not use $\bar{R}^2$? Fixed penalty regardless of sample size. IC is anchored on likelihood theory

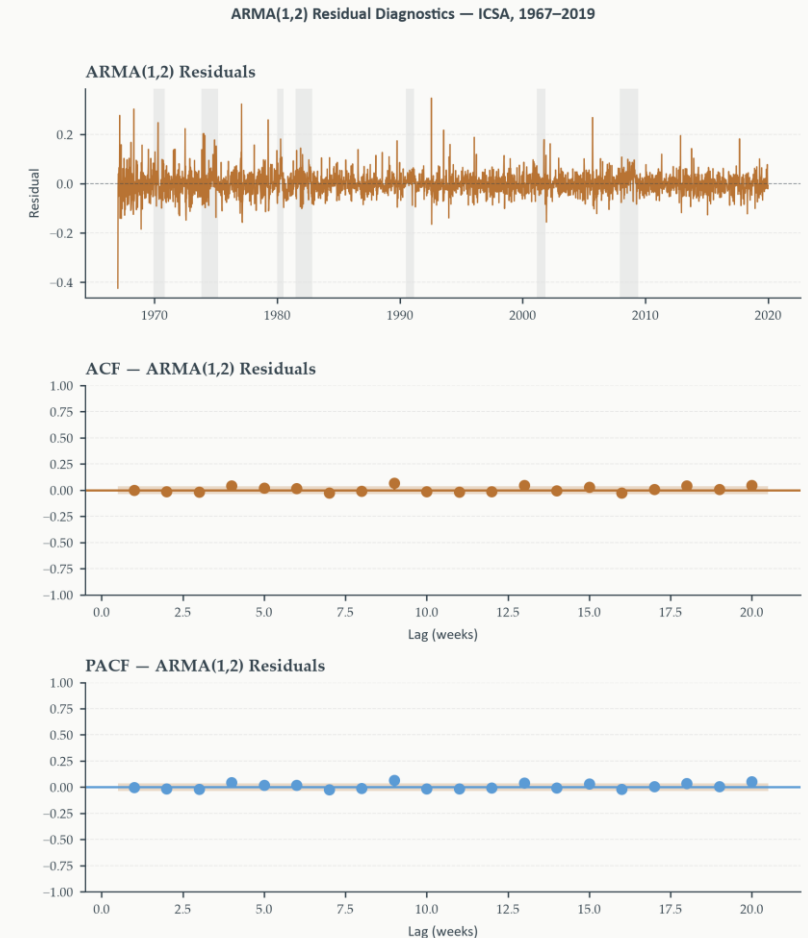
ARMA models and identification

- Fit an AR(1) to unemployment claims
 - $\hat{y}_t = \hat{\phi}_1 y_{t-1} + \hat{c} + \varepsilon_t$
 - Look at the residuals $e_t = y_t - \hat{y}_t$
 - ACF and PACF significant spikes at lags 1 through 3
 - Something is still there...



ARMA models and identification

- Fit an ARMA(1,2) to unemployment claims
 - $\hat{y}_t = \hat{\phi}_1 y_{t-1} + \hat{c} + \varepsilon_t + \hat{\theta}_1 \varepsilon_{t-1} + \hat{\theta}_2 \varepsilon_{t-2}$
 - Look at the residuals $e_t = y_t - \hat{y}_t$
 - ACF and PACF significant spikes at lags 1 through 3
 - Something is still there...



Diagnostics

Section 5

Diagnostics

- A model is a “good fit” if it produces residuals $e \sim WN(0, \sigma^2)$ and $\gamma_e(h > 0) = 0$
- Things to check
 - Is there any serial correlation?
 - Do residuals depict a normal distribution?
 - Are residuals homoscedastic?

Diagnostics

- Serial correlation
 - The Ljung-Box Test
 - H_0 : The first h residual autocorrelations are **jointly** zero
 - $Q(h) = T(T + 2) \sum_{j=1}^h \frac{\hat{\rho}(j)^2}{T-j} \sim \chi_{h-p-q}^2$
 - Estimate $Q(1), Q(2), \dots, Q(h)$ and see where serial correlation cuts off

Be careful! When $T \rightarrow \infty$, the p -values become very small. Always look at the absolute value of $\hat{\rho}(j)$ as well.

Diagnostics

- Normality
 - The Jarque-Bera test
 - H_0 : residuals are normally distributed (based on skewness — S — and kurtosis — K)
 - $JB = \frac{T}{6} \left(\hat{S}^2 + \frac{\hat{K}^2}{4} \right) \sim \chi^2_2$

Diagnostics

- Homoscedasticity
 - Heteroscedasticity test (Harvey test)
 - H_0 : residuals are homoscedastic
 - Let $h = \lfloor T/3 \rfloor$
 - $H(h) = \frac{\sum_{t=T-h+1}^T \bar{v}_t^2}{\sum_{t=d+1}^{d+1+h} \bar{v}_t^2} = \frac{\text{sum of squared residuals in the last third of the sample}}{\text{sum of squared residuals in the first third of the sample}}$
 - $H(h) \sim F(h, h)$
 - $\bar{v}_t^2$ are standardized residuals and d is the number of observations consumed by initial estimation

ARMA models and identification

SARIMAX Results

```
=====
Dep. Variable:      Log Claims    No. Observations:      2765
Model:              ARIMA(1, 0, 2)  Log Likelihood          4592.408
Date:              Thu, 03 Sep 2026  AIC                        -9174.817
Time:              14:58:24        BIC                        -9145.193
Sample:            01-07-1967      HQIC                       -9164.117
                  - 12-28-2019
Covariance Type:    opg
=====
```

	coef	std err	z	P> z	[0.025	0.975]
const	12.6700	0.097	130.683	0.000	12.480	12.860
ar.L1	0.9961	0.002	640.025	0.000	0.993	0.999
ma.L1	-0.3651	0.013	-27.316	0.000	-0.391	-0.339
ma.L2	-0.1376	0.014	-9.603	0.000	-0.166	-0.110
sigma2	0.0021	3.03e-05	69.676	0.000	0.002	0.002

Ljung-Box (L1) (Q):	0.01	Jarque-Bera (JB):	4189.86
Prob(Q):	0.93	Prob(JB):	0.00
Heteroskedasticity (H):	0.45	Skew:	1.01
Prob(H) (two-sided):	0.00	Kurtosis:	8.68

$$\bar{y}^* = \bar{y}(1 - \phi)$$

Warnings:

[1] Covariance matrix calculated using the outer product of gradients (complex-step).

ARMAX

Section 6

ARMAX

- What if past observations of y_t is not enough, what if we need exogenous variables?
 - $\phi_p(L)y_t = \bar{y}^* + \boldsymbol{\beta}'\mathbf{X}_t + \theta_q(L)\varepsilon_t$
 - Why not include past observations of $\mathbf{X}_t$?

ARMAX

- Including exogenous variables opens the model to potential endogeneity bias
 - **When $y_t \leftrightarrow X_t$**
 - A standard solution to this is to move to a VAR(p) model

Forecasting

Section 7

Forecasting

- Point forecasts: ARMA (1,1)
 - **Step 1:** Estimate parameters
 - $\hat{y}_T = \hat{\phi}_1 y_{T-1} + c + \varepsilon_T + \hat{\theta}_1 \hat{\varepsilon}_{T-1}$
 - **Step 2:** Estimate $h = 1$
 - $\hat{y}_{T+1} = \hat{\phi}_1 \hat{y}_T + c + \varepsilon_{T+1} + \hat{\theta}_1 \hat{\varepsilon}_T$
 - **Step 3:** Estimate $h = 2$
 - $\hat{y}_{T+2} = \hat{\phi}_1 \hat{y}_{T+1} + c + \varepsilon_{T+2} + \hat{\theta}_1 \varepsilon_{T+1}$
 - Keep going... you get the mean reversion...
 - $\hat{y}_{T+h \rightarrow \infty} \rightarrow \bar{y}(1 - \hat{\phi}_1)$

Forecasting

- Forecast interval: ARMA (1,1)
 - Express the h -step in its MA(∞) representation:
 - $y_{T+h} = \bar{y} + \varepsilon_{T+h} + \hat{\theta}_1 \varepsilon_{T+h-1} + \hat{\theta}_2 \varepsilon_{T+h-2} + \dots$
 - The h -step forecast error is:
 - $e_h = y_{T+h} - \hat{y}_{T+h} = \varepsilon_{T+h} + \hat{\theta}_1 \varepsilon_{T+h-1} + \hat{\theta}_2 \varepsilon_{T+h-2} + \dots + \hat{\theta}_{h-1} \varepsilon_{T+1}$
 - Only innovation (shocks) from $T + 1$ forward contribute to point estimate uncertainty
 - Take the variance
 - $Var(e_h) = Var[\varepsilon_{T+h} + \hat{\theta}_1 \varepsilon_{T+h-1} + \hat{\theta}_2 \varepsilon_{T+h-2} + \dots + \hat{\theta}_{h-1} \varepsilon_{T+1}]$
 - $Var(e_h) = \sigma^2 \sum_{j=0}^{h-1} \hat{\theta}_j^2, \quad \theta_0 = 1$

Forecasting

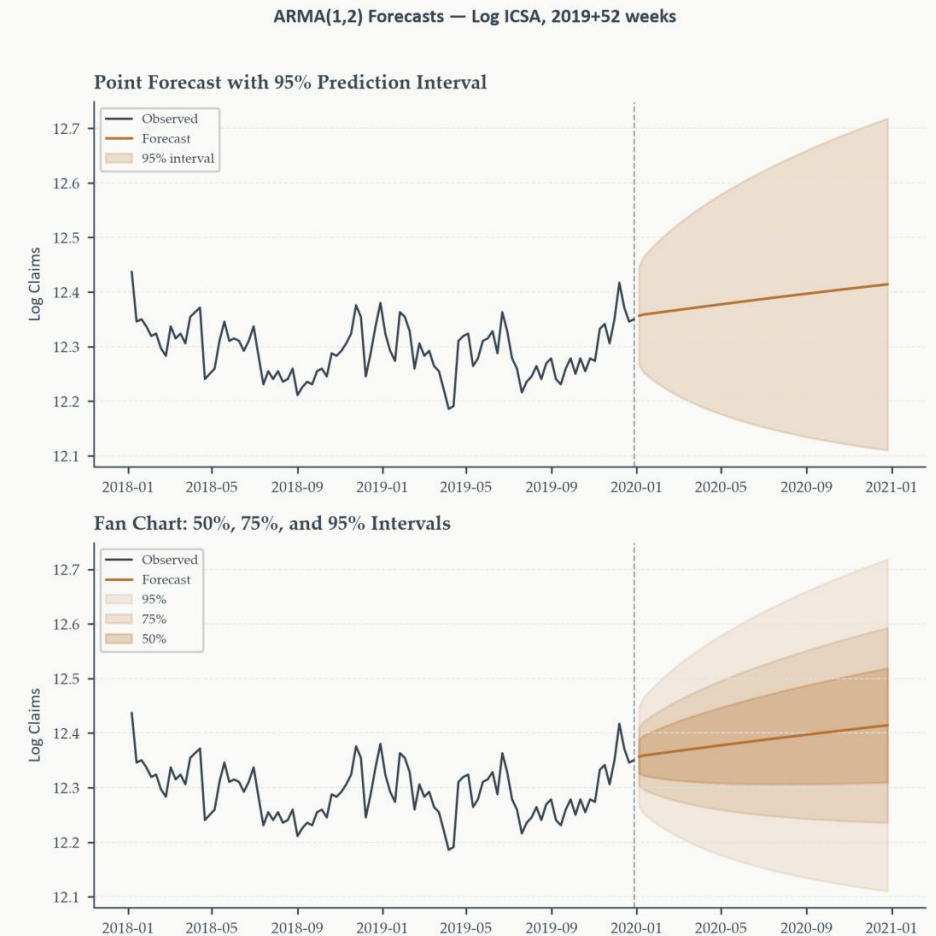
- Forecast interval: ARMA (1,1) (cont...)
 - $Var(e_h) = \sigma^2 \sum_{j=0}^{h-1} \theta_j^2$, $\psi_0 = 1$
 - Note the following:
 - At $h = 1$, the variance is just σ^2
 - For $h > 1$, variance grows at a decreasing rate **if** the series is stationary
 - When $h \rightarrow \infty$, $Var(e_h) \rightarrow \sigma_y^2$

Forecasting

- Forecast interval: ARMA (1,1) (cont...)
 - You need two more things before getting the forecast intervals
 - Probability distribution of $\hat{y}$
 - Confidence level
 - Now you can build your intervals
 - $\hat{y}_{T+h} \pm z_{\frac{\alpha}{2}} \hat{\sigma}_h$

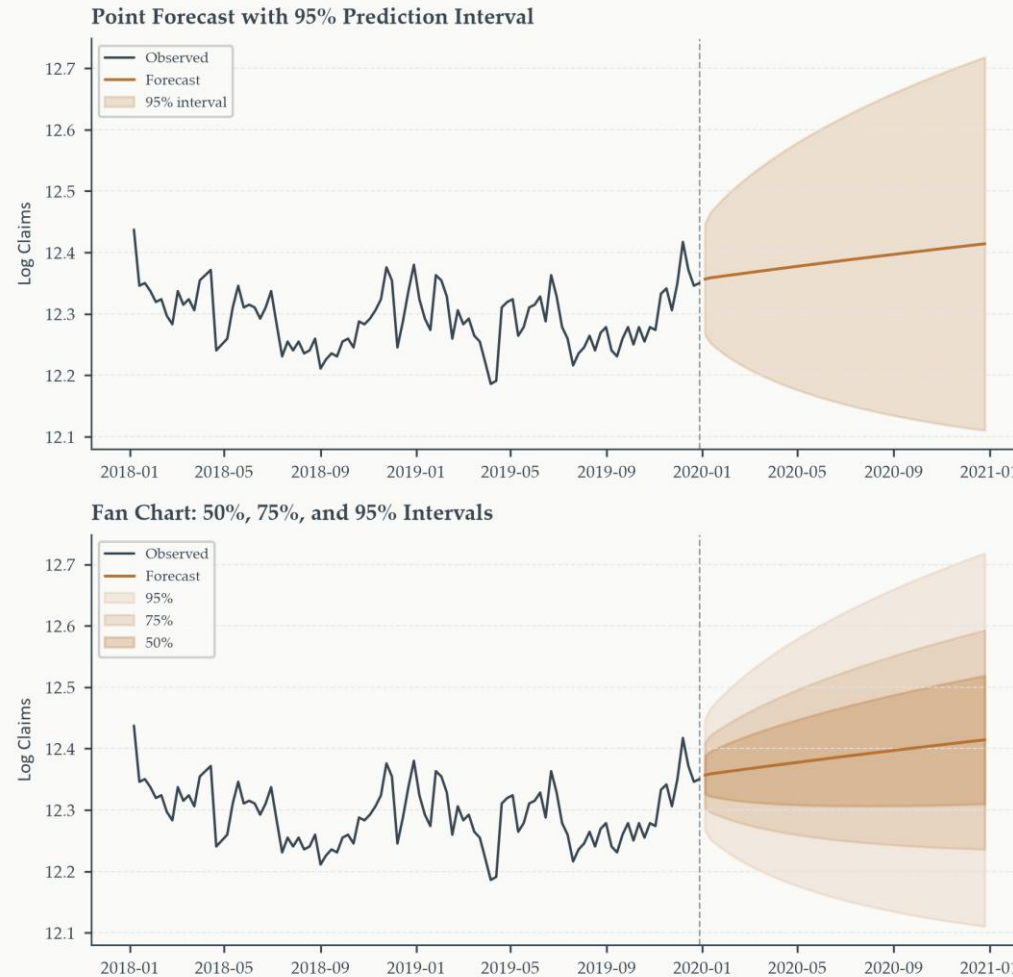
Forecasting

- Fan charts: ARMA (1,1) (cont...)
 - How do you go from a forecast interval to a fan chart?
 - Show multiple confidence intervals simultaneously
 - 50%, 75%, 95%



Forecasting

ARMA(1,2) Forecasts — Log ICSA, 2019+52 weeks



Stationary Time Series Models

Chapter 3

Nicolas Cachanosky

Department of Economics and Finance